



Nigerian Institute of Quantity Surveyors
MEMBERSHIP PORTAL PROGRAMME

API INTEGRATION REQUIREMENTS

Membership Portal — Endpoints Required for Website Integration

The complete list of API routes the Portal team must expose so the NIQS website can integrate member access, directories, payments, jobs and learning into one seamless platform.

PREPARED BY ADLM Studio	FOR Portal Dev Team	VERSION 1.0	DATE 19 Jun 2026
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0 Document Control & Contents

Field	Detail
Document	NIQS Membership Portal — API Integration Requirements
Status	Draft v1.0 — issued for review by the Portal development team
Author / owner	ADLM Studio (NIQS website vendor)
Audience	Membership Portal development team (partner company) · NIQS Secretariat & IT
Purpose	Define every API route, payload and rule the Portal must provide so the two systems integrate into one seamless platform for NIQS members.
Related	Repository docs: <code>PORTAL_INTEGRATION_SPEC.md</code> (low-level contract) and <code>MEMBERSHIP_PORTAL_ACCESS_REQUIREMENTS.md</code> (feature checklist)
Confidentiality	Confidential — shared between NIQS, ADLM Studio and the Portal team only.

How to read this document

The **Portal team builds** the endpoints in Section 3. The **website (ADLM)** consumes them server-to-server with a Bearer API key and never exposes the key to the browser. Each endpoint is tagged **P1** launch-critical, **P2** auth & write-backs, or **P3** future. Start with the P1 set in Section 9.

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1 Background & Ownership Boundary

Two systems make up the NIQS digital platform. The **public website** (built by ADLM Studio) handles public content, events, flyers and the member-facing pages. The **Membership Portal** (portal.niqsng.org, built by your team) is the system of record for member identity, the registered-QS register, dues and member records. For NIQS members to experience one seamless platform, the website must read from — and write a little back to — the portal through a shared API.

1.1 Who owns what

Domain	Owner	Notes
Member identity & login credentials	Portal	Website uses SSO; never stores member passwords.
Membership record (number, type, status, expiry)	Portal	Website reads via API.
Registered-QS directory (individuals)	Portal	Website renders search & segmented views.
Dues, payments, invoices	Portal	Website links to checkout / reads standing.
CPD ledger, exam results, ID card	Portal	Website displays; posts CPD earned at its events.
Public content (news, events, chapters, exco, partners, brand)	Website	—
Events, flyer studio, event registration & attendance/CPD capture	Website	Posts earned CPD back to the portal ledger.
Job board & job applications	Website	Needs member identity (SSO) + optional CV from portal.
QS Firm (company) directory	Website	Today website-managed; could move to portal — see D9.

1.2 Integration channels

Channel	Direction	Use
REST API (server-to-server)	Website → Portal	Verify, profile, directory search, stats, CPD, results, payments. Bearer API key. Phase 1.
SSO / member login	Browser → Portal → Website	Member signs in once against the portal. Phase 2.
Webhooks	Portal → Website	Membership / payment status changes (Section 5). Phase 3.

2 Global API Conventions

These apply to every endpoint in Section 3 so our client needs only configuration, not per-endpoint rewrites.

Item	Convention
Base URL	<code>https://portal.niqsng.org/api/v1</code> (confirm — see D7). One base per environment: staging + production.
Auth	<code>Authorization: Bearer <PORTAL_API_KEY></code> — one key per environment, server-to-server only, never sent to the browser.
Transport	HTTPS only. JSON request & response, UTF-8.
Pagination	List endpoints accept <code>page</code> (1-based) & <code>pageSize</code> ; return <code>total</code> , <code>page</code> , <code>pageSize</code> .
Idempotency	Write-backs (CPD, library) are idempotent on a <code>reference</code> field — retries must not double-count.

2.1 Error envelope

Every non-2xx response uses one shape so both sides parse failures consistently:

```
{ "error": { "code": "string", "message": "human readable" } }
```

200	400	401	403	404	409	422	429
OK	Bad request	Bad / missing key	Not permitted	Not found	Duplicate	Validation	Rate limit

2.2 Shared vocabularies

Keep these identical on both sides (they already match the website's models):

Field	Allowed values
membershipType	<code>probationer</code> · <code>graduate</code> · <code>corporate</code> · <code>fellow</code> · <code>student</code> · <code>technician</code>
status	<code>active</code> · <code>expired</code> · <code>suspended</code> · <code>pending</code>
gender	<code>male</code> · <code>female</code> (required for the WAQSN view — Section 7)
membershipNumber	Canonical key, e.g. <code>NIQS/12345</code> — confirm exact format (D2).

3 Required Endpoints — the Portal builds these

3.1 Member login & Single Sign-On — link to the portal you built

A member clicks **Member Login** on the website and authenticates against **your** portal (single source of truth for credentials). We then mint a short-lived website session. Recommended: redirect + code exchange.

REDIRECT GET /sso/authorize

P2

Browser redirect to your hosted login. On success you redirect back to our callback with a one-time code.

QUERY

```
?client_id=NIQS_WEBSITE &redirect_uri=https://niqs.org.ng/login/portal/callback
&state=<csrf> &scope=profile
```

POST /sso/token

P2

Our server exchanges the one-time code (Bearer API key) for the member + a session handle.

BODY → RESPONSE

```
{ "code": "<one-time>" } → { "member": { ...$3.2 }, "sessionId": "...", "expiresIn": 3600 }
```

If your portal already speaks OIDC, we can consume standard Authorization Code + PKCE instead — tell us which (D1).

POST /auth/introspect

P2

Validate any portal-issued token the browser presents.

BODY → RESPONSE

```
{ "token": "..." } → { "active": true, "member": { ...$3.2 } }
```

3.2 Membership verification

Already wired on our side

Our event-registration "I'm a member" path and the `/membership` verify box already call this through `server/utils/portalClient.js` — they go live the moment this endpoint exists.

Confirm a membership number (or email) belongs to a real, active member.

QUERY

```
?membershipNumber=NIQS/12345 // or ?email=jane@example.com
```

200 RESPONSE

```
{
  "valid": true,
  "member": {
    "membershipNumber": "NIQS/12345", "fullName": "QS Jane Doe",
    "email": "jane@example.com", "membershipType": "corporate",
    "status": "active", "expiresOn": "2026-12-31",
    "chapter": "Lagos", "cpdBalance": 42
  }
} // { "valid": false } when no active member matches
```

3.3 Member profile

Fuller authorised record — powers the member's profile screen in the portal area of the website.

200 RESPONSE

```
{ "member": { ...$3.2 fields, plus
  "phone": "080...", "address": "...", "photoUrl": "https://...",
  "firmName": "Greenfield QS Ltd", "gender": "female",
  "dateOfBirth": "1990-04-12", "yearOfRegistration": 2016 } }
```

3.4 Registered-QS directory search — powers four views from one endpoint

This single endpoint delivers four of NIQS's requested lists

All registered QS, QS in each state chapter, YQSF members and WAQSN (female) members are the same search with different filters. The portal owns the individual register; the website renders the views. (The website's existing `/search-qs-firms` covers **companies** only — the individual register is new and yours.)

Search / list the register with filters. Return only publicly displayable fields (no email/phone in lists).

QUERY PARAMETERS

```
?q=           // name or membership number (free text)
&state=Lagos  // → "QS in this state chapter"
&chapter=lagos // chapter slug or id (see D3)
&type=corporate // membershipType
&status=active // default active
&gender=female // → WAQSN view
&forum=yqsf|waqsn // portal applies the segment rule server-side (§7)
&sort=name|recent &page=1 &pageSize=24
```

200 RESPONSE

```
{
  "results": [
    { "membershipNumber": "NIQS/12345", "fullName": "QS Jane Doe",
      "membershipType": "corporate", "status": "active",
      "state": "Lagos", "chapter": "Lagos", "gender": "female",
      "isYqsf": false, "firmName": "Greenfield QS Ltd", "photoUrl": null }
  ],
  "total": 134, "page": 1, "pageSize": 24
}
```

NIQS list requested	Call
All registered QS (national register)	/members/search (no segment filter)
Registered QS in each state chapter	/members/search?chapter=<slug> or ?state=<state>
YQSF members (Young QS Forum)	/members/search?forum=yqsf
Female members for WAQSN	/members/search?forum=waqsn or ?gender=female

3.5 Directory statistics (counters)

Powers every counter: YQSF "Young Members", per-chapter totals, WAQSN total, homepage figures. Cacheable (e.g. hourly).

200 RESPONSE

```
{
  "total": 10240, "active": 9120,
  "byType": { "corporate": 5400, "graduate": 2600, "fellow": 720, ... },
  "byState": { "Lagos": 1820, "FCT": 940, "Rivers": 510 },
  "byChapter": { "lagos": 1820, "abuja": 940 },
  "female": 2310, "male": 7930,
  "yqsf": 2400, "waqsn": 2310, "updatedAt": "2026-06-19T00:00:00Z"
}
```

3.6 CPD ledger

GET /members/:membershipNumber/cpd

P2

Read the member's lifetime CPD for the portal dashboard tile.

200 RESPONSE

```
{ "balance": 47, "entries": [ { "title": "Digital Cost Management", "points": 5,
  "earnedOn": "2026-08-14", "source": "niqs-website" } ] }
```

POST /members/:membershipNumber/cpd

P2

The website **posts** CPD a member earns by attending a website event. Idempotent on `reference`.

BODY → RESPONSE

```
{ "source": "niqs-website", "eventId": "665f...", "eventTitle": "...",
  "points": 5, "earnedOn": "2026-08-14", "reference": "<our registration id>" }
→ { "ok": true, "cpdId": "...", "newBalance": 47 } // 409 on duplicate = success
```

3.7 Examination results (per member)

Note: the website's `ExamResult` data is only **public sitting announcements** (e.g. "TPC — March 2024, results published June 2024").
Per-member results are yours.

GET /members/:membershipNumber/results

P2

200 RESPONSE

```
{ "results": [ { "examType": "TPC", "sitting": "March 2024",
  "status": "passed", "grade": "-", "resultUrl": "https://..." } ] }
```

3.8 Digital ID card

GET /members/:membershipNumber/id-card

P2

Return a ready card (PDF/image URL) or the fields for the website to render. Confirm which (D8).

200 RESPONSE

```
{ "cardUrl": "https://portal.niqsng.org/cards/NIQS-12345.pdf",
  "validThru": "2026-12-31" }
```

3.9 Payments & member purchases — dues · exams · events · store

The website's `/payment` and `/membership` pages are static placeholders today. The portal owns money.

Phase 1 — hosted checkout deep link (fastest)

LINK /pay?item&membershipNumber&ref

P1

Our "Pay / Renew" buttons deep-link to your hosted checkout, prefilled. After payment, redirect back to us.

```
https://portal.niqsng.org/pay?item=dues&membershipNumber=NIQS/12345&ref=<our-ref>
// on success → redirect to https://niqs.org.ng/payment/success?ref=<our-ref>
```

Current standing (paid / expired) is then read from /members/verify (status , expiresOn).

Phase 2 — payment API (optional, if NIQS wants payment inside the website)

GET /members/:membershipNumber/invoices

P2

200 RESPONSE

```
[ { "id": "inv_1", "item": "Annual dues 2026", "amount": 50000,
    "currency": "NGN", "dueOn": "2026-01-31", "status": "unpaid" } ]
```

POST /payments/checkout

P2

BODY → RESPONSE

```
{ "item": "dues", "membershipNumber": "NIQS/12345", "ref": "<our-ref>" }
→ { "checkoutUrl": "https://portal.niqsng.org/checkout/..." }
```

Plus a payment.succeeded webhook to the website (Section 5). Catalog/price ownership — see D5.

3.10 Job application support

The **job board is website-owned** (search & listings already work). Only the **apply** step needs you: member identity via SSO (§3.1) and, optionally, the member's stored CV so they can "apply with my NIQS CV".

GET /members/:membershipNumber/documents

P2

Optional — only if the portal stores member CVs / documents.

200 RESPONSE

```
[ { "type": "cv", "url": "https://.../cv.pdf", "updatedAt": "2026-05-01" } ]
```

Applications themselves are stored by the website, keyed by membershipNumber (recommended — D6).

3.11 Digital Library — progress tracking (future / "coming soon")

Forward-looking and non-blocking; captured so the contract is ready when the library is built. Member progress must live with the member record.

GET /members/:membershipNumber/library/progress

P3

200 RESPONSE

```
{ "items": [ { "resourceId": "r_88", "title": "BIM for QS",  
  "percent": 100, "completedOn": "2026-09-01", "points": 3 } ] }
```

POST /members/:membershipNumber/library/progress

P3

BODY

```
{ "source": "niqs-website", "resourceId": "r_88", "title": "BIM for QS",  
  "percent": 100, "points": 3, "earnedOn": "2026-09-01", "reference": "..." }
```

If library completion grants CPD, reuse §3.6 instead of a parallel ledger (D10).

4 Endpoint Summary Matrix

#	Method	Path	Purpose	Phase
1	GET	/sso/authorize	Start member login (redirect)	P2
2	POST	/sso/token	Exchange code → member + session	P2
3	POST	/auth/introspect	Validate a portal token	P2
4	GET	/members/verify	Verify membership number / email	P1
5	GET	/members/:no	Full member profile	P1
6	GET	/members/search	Directory — all / state / YQSF / WAQSN	P1
7	GET	/members/stats	Directory counters	P1
8	GET	/members/:no/cpd	Read CPD ledger	P2
9	POST	/members/:no/cpd	Post CPD from website events	P2
10	GET	/members/:no/results	Per-member exam results	P2
11	GET	/members/:no/id-card	Digital ID card	P2
12	LINK	/pay?item&...	Hosted checkout deep link	P1
13	GET	/members/:no/invoices	Member invoices/dues	P2
14	POST	/payments/checkout	Create a checkout session	P2
15	GET	/members/:no/documents	Member CV/documents (jobs)	P2
16	GET	/members/:no/library/progress	Read library progress	P3
17	POST	/members/:no/library/progress	Post library progress	P3

5 Webhooks the Portal calls on the Website

So the website can invalidate sessions/caches and reflect changes in real time. Sign each request with a shared secret: `X-NIQS-Signature: sha256=<hmac(secret, rawBody)>`.

Website endpoint (we build)	Event payload
POST /api/portal/webhooks/membership	{ event: "member.updated member.suspended member.renewed", membershipNumber, status, occurredAt }
POST /api/portal/webhooks/payment	{ event: "payment.succeeded", membershipNumber, item, amount, ref, occurredAt }

For SSO return, the website also exposes `GET /login/portal/callback` (browser) and `POST /api/auth/portal/exchange` (server). We will share exact URLs & the webhook secret during integration.

6 Data Field Mapping (Website ↔ Portal)

Website model	Portal field	Notes
User.membershipId / Registration.membershipNumber	membershipNumber	Canonical key (D2).
membershipType	membershipType	Identical enum (§2.2).
firstName + lastName	fullName	We split/join as needed.
email	email	Lowercased.
Member.chapter (→ Chapter.name/slug/state)	chapter / state	Agree matching key (D3).
Member.gender	gender	Required for WAQSN (§7).
Member.dateOfBirth	dateOfBirth / yearOfRegistration	Needed to compute YQSF (§7).
—	status , expiresOn , cpdBalance	Portal-owned; website reads.
Registration.cpdPoints (per event)	CPD ledger entry	Website posts (§3.6).

7 Segment Rules — YQSF & WAQSN

These business rules must be defined by NIQS and computed by the portal (so the `forum=` filter, the `isYqsf` flag and the counts are consistent everywhere).

YQSF — Young Quantity Surveyors Forum

The website states membership is for QS **under 40 years of age or within their first ten years of professional practice**, and members **auto-exit at 40**. To compute `forum=yqsf` / `isYqsf` / the `yqsf` count, the portal must store **date of birth** and/or **year of first registration/practice**. *Confirm the exact rule (D4).*

WAQSN — Women Association of QS Nigeria

Is WAQSN **all female members** (derive from `gender`), or a **separately enrolled** subgroup some women opt into? If the latter, expose it as `forum=waqsn`; otherwise `gender=female` suffices. Either way the portal must store **gender**. *Confirm (D4).*

8 Security & Operational Requirements

- **API keys** live only in the website server environment — never shipped to the browser. All portal calls are server-side proxied.
- **HTTPS** everywhere; optional **IP allowlist** for server-to-server traffic.
- **Webhooks** signed with a shared secret (HMAC-SHA256 over the raw body).
- **Token TTLs**: SSO session ≤ 1 hour (refresh via re-exchange); one-time codes ≤ 5 minutes.
- **PII minimisation**: directory **search** returns only public fields; email/phone require an authorised profile call.
- **CORS**: allowlist the website origins (production + Vercel preview deployments) for any browser-facing endpoints.
- **Rate limiting + idempotency**: agree [429](#) behaviour + retry/backoff; CPD & library posts idempotent on [reference](#).
- **Environments**: provide **staging** and **production** base URLs and a separate API key for each.

9 Open Decisions & Phased Delivery

9.1 Decisions needed from NIQS / the Portal team

- D1** **SSO style** — redirect + code exchange (recommended) or full OIDC? And is the member area website-rendered or a deep link into your portal UI?
- D2** **Membership-number format** — confirm canonical form (e.g. [NIQS/12345](#)).
- D3** **Chapter matching key** — match members to chapters by chapter id/slug or by state string?
- D4** **YQSF & WAQSN rules** — confirm the YQSF age/practice rule + which fields you store; confirm WAQSN = all female vs opt-in subgroup (§7).
- D5** **Payment catalog** — do prices/SKUs live in the portal (we read them) or stay website-managed item codes?
- D6** **Job applications** — stored website-side (recommended) or posted to the portal? Will you expose member CVs?
- D7** **API base URL** + how keys are issued per environment (staging + production).
- D8** **ID card** — does the portal render the card (return a URL) or return fields for the website to render?
- D9** **QS Firm directory** — stays website-managed or moves to the portal?
- D10** **Digital Library** — content ownership; do completions grant CPD?

9.2 Minimum to unblock launch (priority order)

Order	Deliver	Unlocks
-------	---------	---------

1	Directory search + stats (#6, #7)	All four registered-QS lists + every counter + the /membership verify box
2	Member verify (#4)	Event-registration member path (already wired) + payment standing
3	SSO (#1–3)	Real member login + gates job apply & the dashboard
4	Member profile (#5)	Profile screen in the member area
5	Payments — deep link (#12), then API (#13–14)	Dues/exam/event purchases + "up to date" status
6	CPD / results / ID card / library	Remaining dashboard tiles; library when scheduled

10 What We Need From You — Checklist

10.1 To begin integration

- API **base URLs** — staging & production.
- An **API key** per environment (server-to-server).
- Answers to **decisions D1–D10** (Section 9.1) — especially the YQSF/WAQSN rules, which block those views.
- The **P1 endpoints** on staging first: `/members/search`, `/members/stats`, `/members/verify`.
- For SSO (P2): `client_id`, allowed `redirect_uri`s (production + Vercel previews), token TTLs.
- For webhooks (P3): the shared **HMAC secret**.

10.2 Our side is ready

The website already ships an **env-gated client** (`server/utils/portalClient.js`) implementing `verify` / `profile` / `search` / `firms` / `CPD`. It safely no-ops until configured, so nothing breaks before you go live. When you're ready we set:

```
PORTAL_API_URL=https://portal.niqsng.org/api/v1
PORTAL_API_KEY=*****
PORTAL_SSO_URL=https://portal.niqsng.org/sso/authorize      # Phase 2
PORTAL_OAUTH_CLIENT_ID=NIQS_WEBSITE                       # Phase 2
PORTAL_WEBHOOK_SECRET=*****                              # Phase 3
```

Contact

ADLM Studio — NIQS website team. Please direct API questions, the staging base URL and the API key to the ADLM technical contact. We can join a kick-off call to walk through Sections 3 and 9 together.

End of document · NIQS Membership Portal — API Integration Requirements v1.0 · Prepared by ADLM Studio · 19 June 2026 · Confidential